

Trust by Design

How AI Recommendation Disclosure and Sustainability Framing Shape Consumer Choice in Recommerce Platforms

Document	Research proposal and execution draft
Status	Draft 0.1 — pre-data collection
Research design	Platform audit + pilot + 2 × 2 randomized online experiment
Prepared	13 September 2026

Research integrity note. This draft contains a planned design and literature-grounded hypotheses. It does not report invented findings. Effect sizes, coefficients, significance tests and managerial conclusions should be added only after data collection and analysis.

Executive Summary

Recommerce platforms rely on recommendation systems to reduce search costs and surface products from fragmented inventories. The same platforms also face a trust problem: buyers must judge product condition, seller credibility, authenticity and platform incentives with less certainty than in conventional retail. As AI labels become more visible in digital commerce, the wording around a recommendation may change how consumers interpret both the recommendation and the platform behind it.

This project tests whether disclosing that a product recommendation is AI-generated changes purchase intention on a recommerce platform, and whether the effect depends on the value frame used to justify the recommendation. The proposed experiment crosses AI disclosure with economic versus sustainability framing. The design separates two mechanisms that can move in opposite directions: disclosure may improve perceived transparency, while an AI cue may reduce recommendation credibility when the message asks consumers to accept a more value-laden sustainability claim.

The project uses three stages: a structured audit of public recommerce interfaces to ground the stimuli in current practice; a small pilot to verify the manipulations; and a preregistered 2×2 randomized online experiment with active or recent second-hand shoppers. The primary outcome is purchase intention. Secondary outcomes include recommendation credibility, platform trust, perceived transparency and a simple behavioral proxy such as a click to view the full listing. Analysis will report effect sizes and confidence intervals, use heteroskedasticity-robust regression, and treat mediation as mechanism evidence rather than a definitive causal decomposition.

1. Research Problem and Managerial Context

Recommendation systems matter more in recommerce than in many standard retail settings because inventories are heterogeneous, items are often one-off, and product quality can vary materially across listings. A useful recommendation can reduce search effort, but consumers still need reasons to trust the recommendation and the platform. Prior research on online commerce has shown that trust is closely linked with purchase intention, while second-hand research repeatedly identifies perceived value, risk and trust as central to adoption (Gefen et al., 2003; Kim et al., 2021; Luo & Park, 2024).

AI disclosure adds another layer. Evidence from algorithm research is mixed in a useful way. Consumers sometimes avoid algorithms after observing errors, yet they also rely on algorithmic advice in settings where the task feels objective or quantifiable (Dietvorst et al., 2015; Logg et al., 2019; Castelo et al., 2019). In a field experiment involving more than 6,200 customers, early disclosure that a sales agent was a chatbot sharply reduced purchase rates; more recent transparency studies show that process and performance explanations can raise trust and advice utilization (Luo et al., 2019; Ning et al., 2024). These findings suggest that disclosure design and decision context need to be studied together.

Recommerce provides a particularly useful setting because two common value propositions sit side by side. An economic frame stresses savings, price-quality trade-offs and access to higher-value products. A sustainability frame stresses keeping products in circulation and extending useful life. Both motives are documented in second-hand and collaborative consumption research, and their relative importance varies across consumer segments (Guiot & Roux, 2010; Park & Joyner Armstrong, 2019; White et al., 2019).

2. Literature Review and Research Gap

2.1 AI recommendation disclosure and consumer judgment

AI affects consumer experience through advice, personalization, content generation and automated service. Puntoni et al. (2021) argue that consumer reactions to AI depend on the experience created around the technology, including questions of agency, identity, privacy and social interaction. This perspective is important for recommendation systems because a technically accurate recommendation can still be evaluated negatively if the consumer dislikes how the recommendation was generated or what the source seems to understand about them.

Algorithm aversion and algorithm appreciation provide competing predictions. Dietvorst et al. (2015) show that people can lose confidence in algorithms quickly after seeing them err. Logg et al. (2019) find that people may weight algorithmic advice more heavily than human advice in some judgment tasks. Castelo et al. (2019) reconcile part of this tension by showing that task subjectivity matters: algorithms receive lower trust for tasks that appear subjective, while acceptance improves when the task appears more objective.

Disclosure and transparency add further complexity. Luo et al. (2019) found that revealing chatbot identity before a sales conversation materially reduced purchases, partly through lower perceived knowledge and empathy. Ning et al. (2024) found that performance and process transparency can increase advice utilization through competence beliefs. Park and Yoon (2024) also report that algorithm-transparency signals can build trust in the AI system and transfer to the parent organization. Taken together, the literature supports a two-path account: disclosure can make the system feel more open and accountable, while also making the machine source more salient.

2.2 Recommerce motives, risk and trust

Second-hand consumption combines economic, critical, recreational and product-related motives. Guiot and Roux (2010) developed an eight-factor motivation scale and found meaningful heterogeneity across shoppers. Park and Joyner Armstrong (2019) similarly identify savings, time, assortment and utility motives in online apparel rental and resale. A systematic review by Arrigo (2021) places consumer attitudes, business models and circularity within the same collaborative-fashion research stream.

Trust is especially important when consumers face uncertainty about condition, authenticity or seller claims. Kim et al. (2021) show that additional product-history information can increase trust and perceived benefits in second-hand clothing services. In China, Luo and Park (2024) identify functional, financial, sales-source, physical and psychosocial risks as barriers to second-hand luxury consumption, with perceived value positively associated with purchase intention. These findings support measuring platform trust separately from recommendation credibility.

2.3 Why message framing may change the effect of AI disclosure

Economic framing presents the recommendation as a calculation: the consumer receives a product at a favorable price relative to a credible reference. Sustainability framing asks the consumer to attach environmental or circular-economy value to the same purchase. White, Habib and Hardisty (2019) show that sustainable behavior responds to message design, social influence, identity, affect, cognition and tangibility. Second-hand research also shows that environmental motives coexist with practical and economic motives rather than replacing them.

This difference matters for AI. A price-saving recommendation can be judged using relatively observable information such as price, condition and reference value. A sustainability recommendation carries a stronger normative and identity component. Drawing on task-dependent algorithm aversion, an explicit AI cue may fit an economic recommendation more comfortably than a sustainability claim because the latter invites a more subjective judgment about values and motives (Castelo et al., 2019). Persuasion knowledge offers a second reason to test the interaction: a disclosure can make consumers think more actively about the platform's persuasive intent, especially when the message presents a social or ethical benefit (Friestad & Wright, 1994; Campbell, 1995).

2.4 Research gap and intended contribution

Recent work has examined AI transparency, algorithmic advice and sustainable purchasing, including 2026 studies of recommendation transparency. Much of this evidence uses general e-commerce contexts, perceived transparency measures, or cross-sectional associations. The proposed study adds a controlled recommerce experiment that manipulates the AI disclosure itself and crosses it with the value frame shown on the same product card. This design can identify whether the same disclosure works differently when the recommendation is justified through economic value or sustainability value.

The contribution is deliberately narrow. The study will estimate the causal effect of the two interface cues under a defined product scenario, test whether recommendation credibility and platform trust help explain the pattern, and identify practical communication choices for recommerce platforms. It will not claim to estimate the environmental impact of second-hand consumption or the performance of a live recommendation algorithm.

3. Research Questions and Hypotheses

The literature does not support a confident directional prediction for the total effect of AI disclosure. The study therefore treats the net purchase effect as a research question and preregisters directional hypotheses for mechanisms that have clearer theoretical support.

ID	Statement
----	-----------

- RQ1** What is the total effect of AI recommendation disclosure on purchase intention in a recommerce product decision?

- H1** AI recommendation disclosure will increase perceived transparency relative to a recommendation with no source disclosure.

- H2** Sustainability framing will increase perceived environmental value, while economic framing will increase perceived economic value. These checks establish that the message manipulation changes the intended value interpretation.

- H3** AI disclosure and message framing will interact on recommendation credibility: the credibility penalty associated with AI disclosure will be larger under sustainability framing than under economic framing.

- H4** Recommendation credibility will be positively associated with platform trust, and platform trust will be positively associated with purchase intention.

- H5** The disclosure-by-framing interaction will show an indirect association with purchase intention through recommendation credibility and platform trust. This mechanism test will be reported with bootstrap confidence intervals and interpreted cautiously.

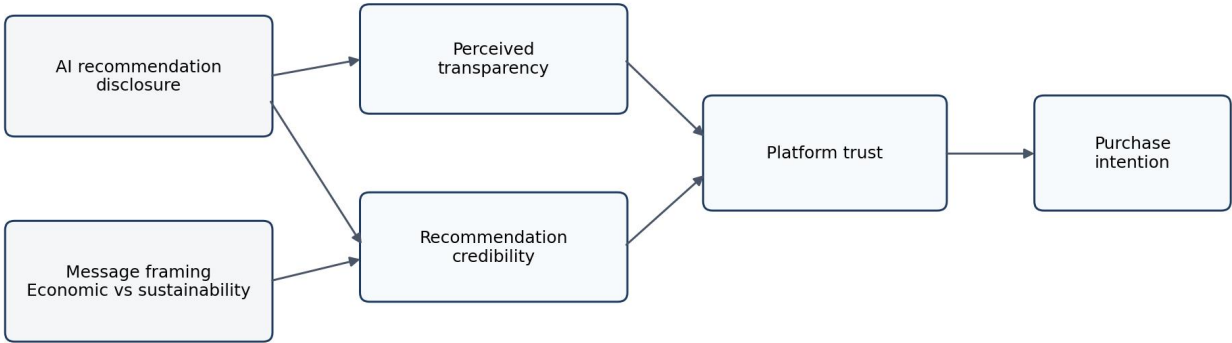


Figure 1. Proposed conceptual model. The main experiment manipulates AI disclosure and value framing; transparency and credibility are measured as distinct mechanisms.

4. Research Design

4.1 Phase A — public platform interface audit

Before building the experiment, the project will audit approximately 90 public product and recommendation interfaces from three major Chinese recommerce platforms, with roughly 30 observations per platform. The purpose is to document how real platforms currently present recommendation cues and value claims so that the experimental stimuli look credible without copying a specific company interface.

The coding sheet will record recommendation-source labels, recommendation rationale, price-saving cues, sustainability cues, authenticity assurance, condition disclosure, seller or platform trust signals, social proof and scarcity cues. A second coder will independently code at least 20% of the sample where feasible. Cohen’s kappa will be reported for categorical variables, with disagreements discussed and the codebook revised before final coding.

4.2 Phase B — pilot test

A pilot with approximately 40–60 eligible participants will test whether the manipulations are visible, believable and similar in perceived design quality. The pilot will also identify wording that creates unintended differences in perceived product quality or price attractiveness. Pilot participants will not enter the main sample.

- AI disclosure check: “It was clear how the platform generated this recommendation.”
- Framing check: separate ratings for economic value and environmental value.
- Realism check: perceived realism of the product page and clarity of the recommendation message.
- Comprehension check: recall of the recommendation label and the key message shown on the product card.

4.3 Phase C — main randomized experiment

The main study uses a 2×2 between-subjects design. Participants are randomly assigned to one of four product-page conditions. All product attributes remain constant across conditions, including item image, price, condition grade, authenticity assurance, seller information, delivery terms and interface layout. Only the recommendation disclosure and value message change.

Condition	Recommendation label	Value message
1. Neutral + Economic	Recommended for you	Save ¥420 compared with the listed new retail price.
2. AI-disclosed + Economic	AI-recommended based on your browsing and saved preferences	Save ¥420 compared with the listed new retail price.
3. Neutral + Sustainability	Recommended for you	Keep this item in use longer by choosing pre-owned.
4. AI-disclosed + Sustainability	AI-recommended based on your browsing and saved preferences	Keep this item in use longer by choosing pre-owned.

A fictitious platform name will be used to avoid prior brand attitudes. The product category should be familiar to young recommerce users and easy to standardize, such as authenticated pre-owned sneakers. The sustainability message will avoid unverified carbon or waste claims. Its wording will focus on extending product use, which can be shown without inventing an environmental impact figure.

4.4 Sample and recruitment

Eligibility: age 18–35; currently living in mainland China; familiar with online shopping; and at least one second-hand platform browse, favorite, inquiry or purchase during the previous 12 months. The study will record prior use of major recommerce platforms but will not require experience with a specific brand.

A practical target is 600 valid responses, approximately 150 per cell, with an initial recruitment target around 660 to allow for preregistered exclusions. Interaction effects require more observations than simple main effects, so the final target should be confirmed with a simulation-based power analysis before launch. If the available recruitment pool cannot support the required sample, the design should be simplified rather than proceeding with an underpowered factorial test.

Recruitment can be conducted through Wenjuanxing distribution channels, university communities and relevant second-hand consumer groups. The final report should disclose the recruitment sources and whether participants received compensation, because sample source can influence external validity.

4.5 Measures

Construct	Operationalization	Role	Literature basis
Purchase intention	3 items; 7-point scale. Likelihood of considering, favoriting or purchasing	Primary outcome	Dodds et al., 1991

the item.

Recommendation credibility	3–4 items covering competence, reliability and trustworthiness of the recommendation.	Key mechanism	Castelo et al., 2019; Ning et al., 2024
Platform trust	3–4 items covering reliability, integrity and confidence in the platform.	Secondary outcome / mediator	Gefen et al., 2003; Delgado-Ballester et al., 2003
Perceived transparency	3 items on clarity of how and why the recommendation was generated.	Mechanism	Ning et al., 2024; Park & Yoon, 2024
Perceived manipulative intent	2–3 items on whether the platform appears to be steering the consumer unfairly.	Alternative mechanism	Campbell, 1995; Friestad & Wright, 1994
Economic value	3 items on savings, value for money and price attractiveness.	Manipulation / mechanism	Guiot & Roux, 2010; Park & Joyner Armstrong, 2019
Environmental value	3 items on product-life extension and perceived environmental benefit.	Manipulation / mechanism	White et al., 2019
Environmental concern	Short validated subset or preregistered brief measure.	Exploratory moderator	Dunlap et al., 2000
AI familiarity	Frequency and confidence using AI-driven recommendations in shopping or digital services.	Exploratory moderator	Study-specific, clearly labelled
Prior recommerce use	Past-12-month browse and purchase frequency.	Precision / heterogeneity	Study-specific behavioral history

All multi-item scales should use consistent 7-point response formats where possible. Items should be adapted carefully to the recommerce context, translated using forward translation and independent back-translation, and pilot-tested for clarity. The final questionnaire should document which items are adapted from prior work and which are original to this project.

4.6 Questionnaire flow

1. Consent, eligibility and prior recommerce use.
2. Random assignment to one of four product-page conditions.
3. Primary outcome measures immediately after exposure, before detailed manipulation checks.
4. Recommendation credibility, platform trust, perceived transparency, perceived manipulative intent, economic value and environmental value.
5. AI familiarity, environmental concern and demographic variables.
6. Attention and manipulation checks, followed by debriefing.

Placing the primary outcome before the manipulation checks reduces the chance that the checks reveal the study purpose and change later answers. A simple behavioral proxy can be added after the main scale items, such as choosing whether to open a full product-detail page. This outcome should be described as a click decision rather than an actual purchase.

5. Analysis Plan

5.1 Preregistration and data exclusions

The hypotheses, sample target, exclusion rules, primary outcome and analysis model should be preregistered before the main study. Exclusions should be limited to objective criteria such as duplicate responses, failed eligibility, impossible completion time, failed instructed-response checks and missing primary outcome data. Condition-specific exclusions after viewing results should be avoided.

5.2 Measurement quality

Internal consistency will be reported using McDonald's omega or Cronbach's alpha, with item-level diagnostics if a scale performs poorly. Construct means will only be formed after reliability checks. If the sample is large enough and the scale set remains compact, confirmatory factor analysis can be used to check whether recommendation credibility, platform trust and transparency are empirically distinguishable.

5.3 Primary factorial model

$$\text{Purchase intention} = \beta_0 + \beta_1 (\text{AI disclosure}) + \beta_2 (\text{Sustainability frame}) + \beta_3 (\text{Disclosure} \times \text{Frame}) + \varepsilon$$

The primary model will be estimated using ordinary least squares with HC3 heteroskedasticity-robust standard errors. Experimental factors will be effect-coded so the interaction is easy to interpret. The report will present cell means, mean differences, 95% confidence intervals and standardized effect sizes. The primary inference will use the unadjusted randomized model; preregistered covariates can be added as a precision check.

Secondary outcomes will use the same factorial structure. Because several related outcomes are tested, the report should distinguish one primary outcome from secondary outcomes and use a correction such as Holm's procedure for the secondary family. Statistical significance should not replace effect-size interpretation.

5.4 Mechanism and heterogeneity analysis

Mechanism analysis will test whether AI disclosure increases perceived transparency and whether the disclosure-by-framing interaction changes recommendation credibility. A serial model can then examine credibility, platform trust and purchase intention. Bootstrap confidence intervals with at least 5,000 resamples can summarize indirect paths. Since mediators are measured after treatment and are not independently randomized, the write-up should describe these results as evidence consistent with a mechanism rather than a fully identified causal mediation effect.

Exploratory heterogeneity will examine prior AI familiarity, environmental concern and recommerce experience. These analyses should be clearly separated from confirmatory tests, use continuous moderators without arbitrary median splits, and report interaction estimates with confidence intervals.

5.5 Robustness checks

- Replicate the primary model with and without preregistered covariates.
- Report results with the full eligible sample and with preregistered attention-check exclusions.
- Check whether product familiarity, perceived product quality or page realism differ across experimental cells.
- For the behavioral click outcome, use logistic regression and report marginal effects.
- Keep the original coding and analysis script in a reproducible R or Python notebook with a data dictionary.

6. Managerial Interpretation Plan

The project should end with a decision rule that a marketing team can use. The result matters because platforms have several ways to disclose AI and several ways to justify a recommendation. The analysis should convert the statistical pattern into an interface and communication recommendation without claiming more than the experiment can show.

Observed pattern	Interpretation	Practical recommendation
Disclosure raises transparency	Consumers reward clear explanation of	Use concise disclosure and explain the

and trust with no purchase penalty	AI involvement.	recommendation basis at the point of recommendation.
Disclosure reduces purchase mainly under sustainability framing	AI source and value-laden sustainability claims appear poorly matched.	Keep AI disclosure, but separate it from broad sustainability persuasion; use verifiable product-life information and stronger evidence.
Disclosure reduces purchase in both frames	Source salience creates resistance that transparency alone does not offset.	Test later-stage disclosure, stronger competence cues, or user control over recommendation settings.
Sustainability framing helps high-concern consumers only	Sustainability is a segmentation variable rather than a universal message.	Personalize message framing based on demonstrated interest or opt-in preferences; keep economic value salient for broader audiences.

Any sustainability recommendation should remain evidence-based. The study should avoid claims about carbon savings, waste reduction or avoided production unless those quantities are independently supported. A defensible recommendation can still be commercially useful: “extend product use” is a narrower claim than “reduce emissions by X%” and is easier for consumers and reviewers to evaluate.

7. Ethics, Data Handling and Research Conduct

- Use only public-facing platform material in the interface audit; do not use confidential employer data, internal dashboards or unpublished company metrics.
- Use a fictitious platform and a research-created product page in the experiment to avoid implying company sponsorship or endorsement.
- Collect informed consent, avoid sensitive personal data, and store only the minimum demographic information needed for analysis.
- Debrief participants that the recommendation and platform were simulated for research purposes. If the study is submitted for academic assessment or publication, follow the university’s ethics-review requirements before data collection.
- Archive the questionnaire, anonymized data, codebook and analysis script where permitted. Record deviations from the preregistered plan in the final report.

8. Anticipated Limitations

The experiment measures a controlled product decision rather than a live purchase with real money. Purchase intention and click behavior can therefore overstate or understate marketplace behavior. A single product category also limits generalization to electronics, luxury goods or household items. The design focuses on disclosure wording and value framing; it does not evaluate recommendation accuracy, model quality or long-term personalization.

The China-focused sample improves relevance for the intended market setting but limits cross-cultural generalization. Sustainability framing can also attract socially desirable responses, which is one reason the design includes a behavioral click proxy and separates environmental value from platform trust. A later replication could add a second product category or an incentive-compatible choice task.

9. Four-Week Execution Plan

Week	Workstream	Output
1	Literature review, 90-interface audit, codebook, first stimulus set	Annotated literature map, coded audit file, four draft product cards

2	Pilot, wording revision, questionnaire translation, preregistration and power simulation	Pilot memo, final stimuli, preregistered analysis plan
3	Main fielding and quality monitoring	Locked raw dataset, response-quality log, data dictionary
4	Analysis, robustness checks, executive interpretation and writing	Research report, reproducible notebook, 8–10 slide strategy deck, one-page framework

10. Final Deliverables

- 12–15 page research report with literature, hypotheses, methods, results, robustness checks and limitations.
- Reproducible R or Python notebook with data dictionary, figures and model outputs.
- 8–10 slide executive strategy deck focused on disclosure design, message framing and customer segmentation.
- One-page “AI Recommendation Trust Framework” translating the experiment into platform design principles.
- Portfolio appendix containing the audit codebook, four stimuli and the final questionnaire.

References

- Arrigo, E. (2021). Collaborative consumption in the fashion industry: A systematic literature review and conceptual framework. *Journal of Cleaner Production*, 325, 129261. <https://doi.org/10.1016/j.jclepro.2021.129261>
- Campbell, M. C. (1995). When attention-getting advertising tactics elicit consumer inferences of manipulative intent: The importance of balancing benefits and investments. *Journal of Consumer Psychology*, 4(3), 225–254. https://doi.org/10.1207/s15327663jcp0403_02
- Castelo, N., Bos, M. W., & Lehmann, D. R. (2019). Task-dependent algorithm aversion. *Journal of Marketing Research*, 56(5), 809–825. <https://doi.org/10.1177/0022243719851788>
- Delgado-Ballester, E., Munuera-Alemán, J. L., & Yagüe-Guillén, M. J. (2003). Development and validation of a brand trust scale. *International Journal of Market Research*, 45(1), 35–53. <https://doi.org/10.1177/147078530304500103>
- Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1), 114–126. <https://doi.org/10.1037/xge0000033>
- Dodds, W. B., Monroe, K. B., & Grewal, D. (1991). Effects of price, brand, and store information on buyers’ product evaluations. *Journal of Marketing Research*, 28(3), 307–319. <https://doi.org/10.1177/002224379102800305>
- Dunlap, R. E., Van Liere, K. D., Mertig, A. G., & Jones, R. E. (2000). New trends in measuring environmental attitudes: Measuring endorsement of the New Ecological Paradigm. *Journal of Social Issues*, 56(3), 425–442. <https://doi.org/10.1111/0022-4537.00176>
- Friestad, M., & Wright, P. (1994). The Persuasion Knowledge Model: How people cope with persuasion attempts. *Journal of Consumer Research*, 21(1), 1–31. <https://doi.org/10.1086/209380>
- Gefen, D., Karahanna, E., & Straub, D. W. (2003). Trust and TAM in online shopping: An integrated model. *MIS Quarterly*, 27(1), 51–90. <https://doi.org/10.2307/30036519>
- Guiot, D., & Roux, D. (2010). A second-hand shoppers’ motivation scale: Antecedents, consequences, and implications for retailers. *Journal of Retailing*, 86(4), 355–371. <https://doi.org/10.1016/j.jretai.2010.08.002>
- Kim, N. L., Woo, H., & Ramkumar, B. (2021). The role of product history in consumer response to online second-hand clothing retail service based on circular fashion. *Journal of Retailing and Consumer Services*, 60, 102457. <https://doi.org/10.1016/j.jretconser.2021.102457>
- Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151, 90–103. <https://doi.org/10.1016/j.obhdp.2018.12.005>
- Luo, H., & Park, S.-Y. (2024). Exploring barriers to secondhand luxury consumption among Chinese consumers and changes during the COVID-19 pandemic. *Journal of Business Research*, 174, 114497. <https://doi.org/10.1016/j.jbusres.2024.114497>
- Luo, X., Tong, S., Fang, Z., & Qu, Z. (2019). Frontiers: Machines vs. humans: The impact of artificial intelligence chatbot disclosure on customer purchases. *Marketing Science*, 38(6), 937–947. <https://doi.org/10.1287/mksc.2019.1192>
- Ning, X., Lu, Y., Li, W., & Gupta, S. (2024). How transparency affects algorithmic advice utilization: The mediating roles of trusting beliefs. *Decision Support Systems*, 183, 114273. <https://doi.org/10.1016/j.dss.2024.114273>
- Park, H., & Joyner Armstrong, C. M. (2019). Is money the biggest driver? Uncovering motives for engaging in online collaborative consumption retail models for apparel. *Journal of Retailing and Consumer Services*, 51, 42–50. <https://doi.org/10.1016/j.jretconser.2019.05.022>

- Park, K., & Yoon, H. Y. (2024). Beyond the code: The impact of AI algorithm transparency signaling on user trust and relational satisfaction. *Public Relations Review*, 50(5), 102507. <https://doi.org/10.1016/j.pubrev.2024.102507>
- Puntoni, S., Reczek, R. W., Giesler, M., & Botti, S. (2021). Consumers and artificial intelligence: An experiential perspective. *Journal of Marketing*, 85(1), 131–151. <https://doi.org/10.1177/0022242920953847>
- White, K., Habib, R., & Hardisty, D. J. (2019). How to SHIFT consumer behaviors to be more sustainable: A literature review and guiding framework. *Journal of Marketing*, 83(3), 22–49. <https://doi.org/10.1177/0022242919825649>

Appendix A. Draft Experimental Stimuli

The four product cards should use identical product images, condition information, authenticity assurance, price, seller rating, delivery terms and layout. The text below is a working draft and should be revised after the pilot.

Element	Draft copy
Neutral recommendation label	Recommended for you
AI disclosure label	AI-recommended based on your browsing and saved preferences
Economic frame	Save ¥420 compared with the listed new retail price.
Sustainability frame	Keep this item in use longer by choosing pre-owned.
Authentication cue	Platform authentication completed. Condition and inspection details available.
Behavioral proxy button	View full listing

The economic comparison must use the same reference price across conditions and should be defensible within the mock scenario. The sustainability copy should avoid precise environmental claims unless a reliable life-cycle source supports them. The AI disclosure should be brief enough to resemble a real interface label rather than a research explanation.

Appendix B. Draft Questionnaire Items

Items below are research-draft wording, informed by the cited constructs and adapted to the recommerce setting. They are intentionally paraphrased. Final wording should be checked against the original scales before fielding and documented in the method section.

Construct	Draft items, 1 = strongly disagree; 7 = strongly agree
Purchase intention	I would seriously consider this item. I would be willing to save or favorite this listing. If I were looking for this type of product, I would consider purchasing it.
Recommendation credibility	This recommendation seems competent. I consider the recommendation reliable. The recommendation gives me confidence that the item is worth considering.
Platform trust	I would feel confident relying on this platform. The platform appears dependable. I believe the platform would handle this recommendation responsibly.
Perceived transparency	The basis of the recommendation is clear to me. I understand why this item was shown to me. The platform is open about how this recommendation was produced.
Manipulative intent	The message feels overly persuasive. I feel the platform is trying to steer my choice too strongly. The recommendation makes me question the platform's motive.
Economic value	This item represents good value for money. The price advantage is meaningful. Buying this item would be financially attractive.

Environmental value

Choosing this item would keep an existing product in use longer.
The purchase would fit with more resource-conscious consumption.
The sustainability message feels relevant to this purchase.

Recommended open-ended question after the scales: “What, if anything, made you trust or distrust the recommendation?” A short coded response can provide useful qualitative context for the statistical results without turning the project into a separate qualitative study.